Maths Homework Template

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Short Math Guide: `http://tug.ctan.org/info/short-math-guide`.

Short Math Guide (Chinese version): `https://github.com/hoganbin/short-math-guide`.

1. ¶ Inline equations for simple math: $\boldsymbol{p} = \hbar\boldsymbol{k}$, $\varepsilon = \hbar\boldsymbol{\omega}$.

   ¶ Centered equations for more involved or important equations:

$$\boldsymbol{a}\cdot\boldsymbol{b} = \sum_{i=1}^{n} a_i b_i \tag{1}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{a(x)} = a'(x)\mathrm{e}^{a(x)} \tag{2}$$

$$\frac{\|A\boldsymbol{x}\|_\infty}{\|\boldsymbol{x}\|_\infty} = \frac{\max\limits_i \left|\sum\limits_{j=1}^{n} A_{ij} x_{j1}\right|}{\max\limits_i |x_{i1}|} \tag{3}$$

   ¶ An example of a matrix:

$$\Psi = \begin{bmatrix} \dfrac{1+\sqrt{3}i}{2} & \mathrm{e}^{-i\omega t} \\ i\mathrm{e}^{-i\omega t} & \dfrac{1-\sqrt{3}i}{2} \end{bmatrix} \tag{4}$$

   ¶ An example of intergals:

$$\iiint \nabla\cdot\boldsymbol{A}\,\mathrm{d}V = \oiint \boldsymbol{A}\,\mathrm{d}\boldsymbol{S} \tag{5}$$

   ¶ An example of roots:

$$r = \sqrt[3]{\frac{3}{4\pi N}} \tag{6}$$

2.

$$\begin{aligned} I(x) &= \int_{-\frac{b}{2}}^{\frac{b}{2}} \mathrm{d}I(x) \\ &= 4\alpha\frac{I_0}{b}\int_{-\frac{b}{2}}^{\frac{b}{2}} \cos^2\left(\frac{\pi d}{\lambda D}\left(x + \frac{D}{R}\xi\right)\right)\mathrm{d}\xi \\ &= 4\alpha I_0\left(1 + \frac{1}{u}\sin u\cos\left(\frac{2\pi d}{\lambda D}x\right)\right), \quad u = \frac{\pi d b}{\lambda R} \end{aligned}$$

3.

$$Y_n(y) = \begin{cases} 0, & n = 2k \\ \dfrac{8a^2}{\pi^3 n^3}\left(1 - \dfrac{\cosh\dfrac{n\pi y}{a}}{\cosh\dfrac{n\pi b}{2a}}\right), & n = 2k+1 \end{cases}$$

4.

$$
\begin{array}{cccc}
\mathcal{ABCD} & \mathscr{ABCD} & \boldsymbol{\mathfrak{ABCD}} & \mathbb{ABCD} \\
\mathsf{\mathit{ABCD}} & \boldsymbol{ABCD} & \mathrm{ABCD} & ABCD
\end{array}
$$

5.

$$
u(x,t)=\sum_{n=0}^{\infty}X_n(x)Y_n(y)=\frac{8a^2}{\pi^3}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^3}\left(1-\frac{\cosh\dfrac{2k+1}{a}\pi y}{\cosh\dfrac{2k+1}{2a}\pi b}\right)\sin\frac{2k+1}{a}\pi x
$$

6.

$$
\begin{aligned}
\mathscr{L}\left\{\frac{1-\cos\omega t}{t}\right\} &= \int_p^{\infty}\left(\frac{1}{p}-\frac{1}{2(p+\mathrm{i}\omega)}-\frac{1}{2(p-\mathrm{i}\omega)}\right)\mathrm{d}t \\
&= \ln p-\frac{1}{2}\ln(p+\mathrm{i}\omega)-\frac{1}{2}\ln(p-\mathrm{i}\omega)\bigg|_p^{\infty} \\
&= \frac{1}{2}\ln\frac{p^2+\omega^2}{p^2}
\end{aligned}
$$

7.

$$
\begin{aligned}
[q,p]_{Q,P} &= \frac{\partial q}{\partial Q}\cdot\frac{\partial p}{\partial P}-\frac{\partial q}{\partial P}\cdot\frac{\partial p}{\partial Q} \\
&= 1+\frac{P^2\mathrm{e}^{2Q}}{1-P^2\mathrm{e}^{2Q}}-\frac{P^2\mathrm{e}^{2Q}}{1-P^2\mathrm{e}^{2Q}} \\
&= 1
\end{aligned}
$$

8.

$$
\begin{aligned}
\frac{4\pi R_\odot^2\cdot\sigma T_\odot^4}{4\pi d^2} &= 1352.83\,\mathrm{W/m^2} \\
T_\odot &= 5763.43\,\mathrm{K} \\
\lambda_\odot = \frac{b}{T_\odot} &= 5027.86\,\text{Å}
\end{aligned}
$$